Response to Editor and Reviewers

Manuscript: "JDCNet Uses Confidence Gated Privileged Modality Distillation for Cost Preserving X Ray Inference"
Journal: Discover Artificial Intelligence
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We thank the Editor and Reviewers for their careful second-round assessment. We have revised the manuscript and address every point below. Section, table, and figure references refer to the revised clean manuscript. No tracked changes are included in the manuscript upload.

Editorial requirements

1. Revise the title so that it has a clear scientific meaning and no punctuation. The title is now "JDCNet Uses Confidence Gated Privileged Modality Distillation for Cost Preserving X Ray Inference". It is used identically in the manuscript, the separate title-page source, and the cover letter; punctuation and the former hyphenated subtitle structure have been removed.

2. Select Yes for data availability and make the statement match the manuscript. The submission statement is now set to "Yes. I used or generated research data in this study." The manuscript and the editorial-system statement now point to the BIMCV-COVID19+ portal at https://bimcv.cipf.es/bimcv-projects/bimcv-covid19 and explicitly identify the BIMCV-COVID19 Research Use Agreement at https://bimcv.cipf.es/bimcv-projects/bimcv-covid19/bimcv-covid19-dataset-research-use-agreement-2/. The statement records that the data are de-identified, that re-identification and data sharing are prohibited, and that no image pixels or individual image files are redistributed. Only derived aggregate results and anonymized case-level gate summaries are reported; split definitions, aggregate results, and analysis manifests are available at https://github.com/mabo1215/JDCNET.git.

3. Add a Declarations section with separate ethical approval, consent to participate, and consent to publish headings. Done. The manuscript now contains a separate *Declarations* section with individual *Ethical approval*, *Consent to participate*, and *Consent to publish* subheadings, in addition to funding, competing interests, data availability, materials availability, code availability, and author contribution statements. The study is a secondary analysis of publicly released de-identified data under the BIMCV Research Use Agreement, with no new recruitment or data collection. That agreement permits research use but prohibits sharing, re-identification, and publication or reproduction of any dataset portion without specific prior written permission. We therefore provide the official BIMCV data-portal URL, not patient image files or pixels.

Reviewer 1

We thank the Reviewer for confirming that the requested corrections were addressed.

Reviewer 3

1. Clarify how the framework will be validated on larger multi-institutional and multi--disease datasets. We agree. The Limitations and Future Work section now specifies a validation programme rather than only a general aspiration. The next study will assemble same-patient CT–X-ray cohorts from multiple hospitals, use site-held-out evaluation, and harmonize label definitions before extending the task from COVID-19 versus non-COVID to multi-disease classification (for example, pneumonia, tuberculosis, edema, and no-finding categories). The transfer gate, paired bootstrap, and per-fold win-count reporting will be fixed before those data are evaluated. These are explicitly described as planned validation steps, not as results established by the present single-cohort study.

2. Add qualitative visualizations of accepted and rejected teacher predictions or attention maps. We added a new qualitative gate-behaviour figure, an aggregate qualitative visualization of the gate behavior. It contrasts the retained subset (ECE = 0.07–0.08 and retained-minus-rejected teacher accuracy of +0.11–+0.14) with the rejected subset (ECE = 0.14–0.16), and shows the DRR failure mode in which confidence does not identify a reliable subset. The underlying per-teacher coverage and reliability values remain in the gate-coverage table. JDCNet uses a ResNet-18 backbone and contains no attention module; attention maps would therefore add an unvalidated interpretation rather than visualize a mechanism used by the method. We label the new panel as an aggregate diagnostic, not as an individual-patient explanation. To answer the request for real-sample context while respecting the BIMCV Research Use Agreement, we added a four-row case-level numerical table to the revised manuscript, generated from the retained mid-slice teacher checkpoint, the fold-0/seed-42 manifest, and the available de-identified CT cache. It reports teacher confidence, teacher prediction, ground truth, and gate outcome at $\tau_{\text{gate}} = 0.80$. The examples include accepted and rejected predictions, including an accepted incorrect prediction and a rejected incorrect prediction, so that acceptance is visibly distinct from correctness. The table uses newly assigned case labels, cites the official BIMCV data portal as the source collection, and does not reproduce pixels, individual image URLs, filenames, or patient identifiers. It is explicitly labelled as illustrative context rather than a new performance estimate.

3. Explain the practical or therapeutic implications of the balanced-accuracy gains. We added a practical interpretation immediately after the absolute-performance table. For the strongest 3-slice soft-KL cell, balanced accuracy increases from 0.604 to 0.639, sensitivity from 0.547 to 0.598, and specificity from 0.661 to 0.679. This indicates why the result could matter for X-ray triage or prioritisation: more positive cases are detected at the reported balanced operating point without a specificity decrease in this cohort. We also state explicitly that these are retrospective model metrics, not evidence of improved treatment decisions, patient outcomes, or therapeutic benefit; those claims require prospective workflow and outcome studies.

4. Discuss recently published lightweight medical-imaging models. The Related Work section now positions MobileNetV2 and EfficientNet as relevant lightweight deployment backbones. The Deployment-Time Efficiency section also reports a portability check using MobileNetV2 and EfficientNet-B0 students with the same gated supervision: the observed gains are positive but slightly below the pre-specified gate (+0.026 and +0.029, respectively). This strengthens the practical discussion while keeping the claim bounded: the validated headline deployment path remains the ResNet-18 student, and the lightweight results are portability evidence rather than independent clinical validation.

5. Expand the external-validation discussion with domain adaptation and calibration options. The Limitations section now separates acquisition-protocol shift, label-definition/prevalence shift, and population/scanner shift as plausible causes of the external collapse. It discusses multi-center training, unsupervised or lightly supervised feature/input domain adaptation, and post-hoc site-specific threshold recalibration as concrete next experiments. The manuscript also reports the teacher-temperature ablation in the Appendix. We make clear that none of the proposed mitigations has been validated in this paper and that the external X-ray stress test does not replace independent same-patient paired replication.

Reviewer 4

1. Make the novelty relative to confidence-aware KD and privileged-information methods explicit. We added the Methodology subsection "Design Rationale versus Confidence-Aware and Confidence-Based Distillation" and the Related Work comparison in the SOTA comparison table. The revised text distinguishes JDCNet along two axes: a hard per-sample confidence mask rather than continuous reliability weighting, and a discrete or lightly softened auxiliary target rather than a fully softened cross-modal logit distribution. It also compares modality pairing, same-patient evaluation, fixed-cost deployment auditing, target granularity, and failed-channel reporting against recent privileged-transfer and confidence-aware approaches. The claim is stated as a joint methodological and evaluation contribution, not as a claim that confidence gating itself is new in every domain.

2. Compress repetitive material in the Introduction, Related Work, and Discussion. We shortened the repeated positioning paragraph in the Introduction and condensed the Discussion's broad clinical-imaging implications so that they refer back to the design-rationale subsection rather than restating the full mechanistic argument. Related Work now uses the comparison table to carry the repeated method-by-method distinctions, while retaining only the differences needed to establish the scope and novelty of JDCNet. The revision preserves the evidence limitations and citations while removing duplicated cost and cross-modal-transfer explanations.

3. Add implementation details for seeds, preprocessing, hyperparameter selection, and computer resources. The Implementation Details section specifies ImageNet-pretrained ResNet-18 initialization, seeds 42–44, patient-level five-fold cross-validation, the 224×224 resize, augmentation, channel replication for CT slices, weighted sampling, weighted cross-entropy, AdamW settings, 50 epochs, the pre-specified τ and λ grid, and the RTX 3090 environment. It also reports the approximately six-hour two-GPU sweep, the approximately $1.18\times$ training-time overhead, and the measured deployment cost on CPU, GPU, and edge hardware. The Appendix gives the cohort construction and leakage audit.

4. Investigate the performance drop on the independent dataset. We expanded the External-Cohort Stress Test and Limitations sections. The frozen supervised and JDCNet students all fall to chance on the external X-ray-only cohort, so the result is reported as a source-to-target generalization warning rather than a JDCNet-specific failure. We now discuss acquisition, population, scanner, label, and prevalence shift and identify multi-center training, domain adaptation, and site-specific calibration as follow-up approaches. The manuscript does not present any of these approaches as already validated.

5. Add stronger statistical comparisons. The main decision-summary table now reports the paired ΔBA, bootstrap confidence interval, Wald-type z, two-sided p, and paired Cohen's d for the headline rows and strongest competing baselines. The primary decision remains the pre-specified CI-based transfer gate with Benjamini–Hochberg correction; the named tests and effect sizes are reported as complementary diagnostics. Both passing JDCNet settings remain significant ($p = 0.003$ and $p = 0.011$), while the strongest failed gated-logit comparator is not significant ($p = 0.19$).

6. Expand clinical significance, workflow integration, regulatory issues, and real-world limitations. The revised manuscript interprets the absolute sensitivity and specificity changes, explains that CT is removed before deployment, and discusses how an X-ray-only input path could fit a radiology workflow without requiring CT at inference. It also lists the unmet requirements before clinical use: multi-site external validation, prospective outcome-based evaluation, site-specific threshold calibration, and regulatory assessment of the deployed X-ray decision-support software. The benefits-and-limitations table summarizes these benefits, limitations, and implementation risks in one place.

7. Polish the language and summarize the main advantages and disadvantages. We performed a language pass over the Introduction, Methods, Results, and Discussion, removing repeated claims and replacing broad deployment language with evidence-bounded wording. The benefits-and-limitations table now provides the requested compact summary: zero inference-time CT cost and the observed balanced-accuracy gain are listed alongside training overhead, dependence on a paired single-cohort dataset, lack of prospective clinical evidence, and outstanding regulatory and calibration requirements.

Summary of files changed

The clean manuscript source, appendix source, bibliography, figures, title-page source, data-availability statement, cover letter, and this point-by-point response have been updated. The revised clean manuscript and response have been compiled and are supplied as PDF files for upload. A marked-up manuscript is optional under the Editor's instructions and is not included in the clean manuscript upload.